In [3]:
```
import sys
!{sys.executable} -m pip install torch torchvision torchaudio
```

```
Collecting torch
  Downloading torch-2.8.0-cp39-cp39-win_amd64.whl (241.2 MB)
     ------------------------------------ 241.2/241.2 MB 981.2 kB/s eta 0:00:00
Collecting torchvision
  Downloading torchvision-0.23.0-cp39-cp39-win_amd64.whl (1.6 MB)
     ---------------------------------------- 1.6/1.6 MB 1.1 MB/s eta 0:00:00
Collecting torchaudio
  Downloading torchaudio-2.8.0-cp39-cp39-win_amd64.whl (2.5 MB)
     ---------------------------------------- 2.5/2.5 MB 2.6 MB/s eta 0:00:00
Collecting sympy>=1.13.3
  Downloading sympy-1.14.0-py3-none-any.whl (6.3 MB)
     ---------------------------------------- 6.3/6.3 MB 3.7 MB/s eta 0:00:00
Requirement already satisfied: networkx in c:\users\admin\anaconda3\lib\site-packages
(from torch) (2.8.4)
Requirement already satisfied: fsspec in c:\users\admin\anaconda3\lib\site-packages
(from torch) (2022.7.1)
Requirement already satisfied: filelock in c:\users\admin\anaconda3\lib\site-packages
(from torch) (3.6.0)
Requirement already satisfied: jinja2 in c:\users\admin\anaconda3\lib\site-packages
(from torch) (2.11.3)
Collecting typing-extensions>=4.10.0
  Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
     ---------------------------------------- 44.6/44.6 kB 2.1 MB/s eta 0:00:00
Requirement already satisfied: numpy in c:\users\admin\anaconda3\lib\site-packages (f
rom torchvision) (1.21.5)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in c:\users\admin\anaconda3\lib
\site-packages (from torchvision) (9.2.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\users\admin\anaconda3\lib\sit
e-packages (from sympy>=1.13.3->torch) (1.2.1)
Requirement already satisfied: MarkupSafe>=0.23 in c:\users\admin\anaconda3\lib\site-
packages (from jinja2->torch) (2.0.1)
Installing collected packages: typing-extensions, sympy, torch, torchvision, torchaud
io
  Attempting uninstall: typing-extensions
    Found existing installation: typing_extensions 4.3.0
    Uninstalling typing_extensions-4.3.0:
      Successfully uninstalled typing_extensions-4.3.0
  Attempting uninstall: sympy
    Found existing installation: sympy 1.10.1
    Uninstalling sympy-1.10.1:
      Successfully uninstalled sympy-1.10.1
Successfully installed sympy-1.14.0 torch-2.8.0 torchaudio-2.8.0 torchvision-0.23.0 t
yping-extensions-4.15.0
```

```
WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=Non
e)) after connection broken by 'ReadTimeoutError("HTTPSConnectionPool(host='pypi.or
g', port=443): Read timed out. (read timeout=15)")': /simple/torch/
WARNING: Retrying (Retry(total=3, connect=None, read=None, redirect=None, status=Non
e)) after connection broken by 'ProtocolError('Connection aborted.', ConnectionResetE
rror(10054, 'An existing connection was forcibly closed by the remote host', None, 10
054, None))': /simple/torch/
WARNING: Retrying (Retry(total=2, connect=None, read=None, redirect=None, status=Non
e)) after connection broken by 'ProtocolError('Connection aborted.', ConnectionResetE
rror(10054, 'An existing connection was forcibly closed by the remote host', None, 10
054, None))': /simple/torch/
  WARNING: Retrying (Retry(total=4, connect=None, read=None, redirect=None, status=No
ne)) after connection broken by 'ProtocolError('Connection aborted.', ConnectionReset
Error(10054, 'An existing connection was forcibly closed by the remote host', None, 1
0054, None))': /packages/b9/dc/1f1f621afe15e3c496e1e8f94f8903f75f87e7d642d5a985e92210
cc208d/torch-2.8.0-cp39-cp39-win_amd64.whl
```

In [4]:
```python
import torch
print(torch.__version__)
```

```
2.8.0+cpu
```

In [5]:
```python
import torch
import torch.nn as nn
import math
```

In [6]:
```python
# Positional Encoding
class PositionalEncoding(nn.Module):
    def __init__(self, d_model, max_len=5000):
        super().__init__()

        pe = torch.zeros(max_len, d_model)
        position = torch.arange(0, max_len).unsqueeze(1)

        div_term = torch.exp(torch.arange(0, d_model, 2) *
                             (-math.log(10000.0) / d_model))

        pe[:, 0::2] = torch.sin(position * div_term)
        pe[:, 1::2] = torch.cos(position * div_term)

        pe = pe.unsqueeze(0)
        self.register_buffer('pe', pe)

    def forward(self, x):
        return x + self.pe[:, :x.size(1)]
```

In [7]:
```python
# Simple Transformer Model
class SimpleTransformer(nn.Module):
    def __init__(self, vocab_size, d_model=128, nhead=4, num_layers=2):
        super().__init__()

        self.embedding = nn.Embedding(vocab_size, d_model)
        self.pos_encoder = PositionalEncoding(d_model)

        encoder_layer = nn.TransformerEncoderLayer(
        d_model=d_model,
        nhead=nhead,
        batch_first=True
        )
```

```
        self.transformer_encoder = nn.TransformerEncoder(
            encoder_layer,
            num_layers=num_layers
        )

        self.fc_out = nn.Linear(d_model, vocab_size)

    def forward(self, x):
        x = self.embedding(x)
        x = self.pos_encoder(x)

        x = self.transformer_encoder(x)

        x = self.fc_out(x)

        return x


# Testing the Transformer
vocab_size = 1000
seq_len = 10
batch_size = 2

model = SimpleTransformer(vocab_size)

sample_input = torch.randint(0, vocab_size, (batch_size, seq_len))

output = model(sample_input)

print("Input Shape:", sample_input.shape)
print("Output Shape:", output.shape)
```

```
Input Shape: torch.Size([2, 10])
Output Shape: torch.Size([2, 10, 1000])
```

In [ ]: